NEUROSANBE

Film-coated tablet

COMPOSITION
Each film-coated tablet contains:
Vitamin B_1 .... 100 mg
(equivalent with 111.047 mg Thiamine Mononitrate)
Vitamin B_6 .... 200 mg
(equivalent with 200 mg Pyridoxine Hydrochloride)
Vitamin B_{12} .... 200 mcg

Specification: **In-house**

PRODUCT DESCRIPTION
Tablet, biconvex, tasteless, breakline and markings SB, and with rose color film coated tablet.

PHARMACOLOGY
NEUROSANBE is a combination of three essential neurotropic vitamins (B1, B6 and B12) in high dosage. Vitamins B1, B6 and B12 are of special importance for the metabolism in the peripheral and central nervous system. Their effect on the regeneration of nerves has been shown in various investigations using the vitamins individually and in combination.

Vitamin B1 plays an important role in major metabolic processes. Vitamin B6 has an analgesic effect. Vitamin B12 ensures blood cell formation and prevents degenerative processes of the nervous system. Both the individual function and the beneficial biochemical links between the three vitamins justify their combined use.

PHARMACOKINETICS
The combined administration of the vitamins B1, B6 and B12 has no effect on the pharmacokinetics of the individual vitamins.
Thiamine (vitamin B1):
Shows a dose-related dual transport mechanism following oral dosing: Active absorption up to concentrations of 2 µmol and passive diffusion for concentrations in excess of 2 µmol.

Studies using radiolabeled thiamine showed that duodenal absorption is highest, while absorption in the upper and medium small bowel segments is somewhat lower. Hardly any absorption is seen in the stomach and in the distal segments of the small bowel. Thiamine being formed in the small bowel is not absorbed.

Following absorption by the intestinal mucosa thiamine is transported into the liver via the hepatic portal system. In the liver, thiamine is phosphorylated to thiamine pyrophosphate (TPP) and thiamine triphosphate (TTP) by thiamine kinase.
The elimination half-life of thiamine is 1 hour. The primary excretion products are: thiamine carboxylic acid, pyramin, thiamine and some other metabolites not yet identified (renal excretion). The amount of unmetabolised thiamine eliminated by renal route within 4-6 hours will increase with increasing thiamine uptake.
Pyridoxine (vitamin B6):
Vitamin B6 (pyridoxine, pyridoxal and pyridoxamin) is rapidly absorbed predominantly in the upper gastrointestinal tract and transported to the organs and into tissue. Vitamins are bound to albumin. Approximately 80% of pyridoxal phosphate is bound to proteins. Vitamin B6 passes into spinal fluid, is excreted in breast milk and crosses the placenta barrier. The main elimination product is 4-pyridoxine acid with the amount depending on the vitamin B6 dose administered.
Cyanocobalamin (vitamin B12):
Absorption from the gastrointestinal tract is based on 2 different mechanisms:
- release by gastric acid and immediate binding to the intrinsic factor forming the vitamin B12 intrinsic factor complex.
- independent of the intrinsic factor by passive influx into blood.

When oral dose is increased, the intrinsic-factor-related absorption will reach saturation and diffusion-induced absorption will increase. Approximately 90% of plasma cobalamin is bound to proteins (transcobalamins). The major part of the non-circulating vitamin B12 in plasma is stored in the liver. Vitamin B12 is predominantly secreted into bile and largely reabsorbed by enterohepatic circulation. If the storage capacity of the body is exceeded as a result of high doses and, in particular, parenteral administration, the excessive share will be eliminated in the urine.

INDICATION
Diabetic polyneuropathy, alcoholic peripheral neuritis, post-influenzal neuropathies, neuritis & neuralgia of the spinal nerves especially facial paresis, cervical syndrome, low back pain, ischialgia.

CONTRA-INDICATION
Hypersensitivity to any of the active ingredients or excipients of **NEUROSANBE**. Use in children: **NEUROSANBE** film-coated tablets are not suitable for the treatment of children due to the high content of vitamin B-complex.

ADVERSE REACTION
Hypersensitivity reactions to vitamin B_1 e.g., sweating, tachycardia (rapid heart beat), and skin reactions with itching and urticaria are very rare. Gastrointestinal complaints e.g., nausea, vomiting, diarrhoea or abdominal pain may occur.

PRECAUTIONS
Neuropathies are described under long-term intake (6-12 months) of > 50 mg mean daily dose of vitamin B_6. Therefore, under long-term treatment, regular monitoring is recommended. **NEUROSANBE** film-coated tablets contain lactose and sucrose; therefore, its use is not recommended in patients with intolerance to some sugars (i.e., rare hereditary galactose or fructose intolerance, glucose-galactose malabsorption, Lapp lactase deficiency, or sucrose-isomaltase insufficiency).

PREGNANCY & LACTATION
No risks have become known associated with the use of **NEUROSANBE** during pregnancy at the recommended dosage. Vitamins B1, B6 and B12 are secreted into human breast milk, but risks of over dosage for the infant are not known. In individual cases, high doses of vitamin B6 i.e., > 600 mg daily, may inhibit the production of breast milk.

DRUG INTERACTION
The effect of Levodopa may be reduced when vitamin B_6 is administered concomitantly. Pyridoxine-Antagonists e.g., INH, Cycloserine, Penicillamine, Hydralazine: The efficacy of vitamin B_6 (pyridoxine) may be decreased. Loop Diuretics e.g., Furosemide: In long-term use, the blood level of thiamine may be reduced.

DOSAGES
1 coated tablet three times daily to treat moderate cases, or to provide interval and follow-up therapy for a course of injections.
NEUROSANBE coated tablets are swallowed without chewing with a little liquid with or after meals.
The duration of the treatment is determined by the doctor.

OVER DOSAGES
Prolonged overdose of vitamin B6 i.e., for > 2 months and > 1 g/day, may lead to neurotoxic effects.

PRESENTATION
Box of 10 strips @ 10 tablets, and 50 strips @10 tablets.
MAL No.: MAL14015022XZ

ROUTE OF ADMINISTRATION
Oral administration.
Should be taken with food. Take with or after meals. Swallow whole, do not chew/crush.

STORAGE
Store at temperature below 30°C.

KEEP OUT OF REACH OF CHILDREN

JAUHI DARIPADA KANAK-KANAK

Manufactured by:
PT SANBE FARMA
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